A Blockchain-Enabled Framework for Misinformation Monitoring

Multilingual AI triage and a shared, tamper-evident ledger for South Africa's election fact-checkers

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Introduction

In South Africa's 2024 election, AI deepfakes spread through politically active WhatsApp groups, and the fact-checks posted in reply were often dismissed as propaganda.

Fact-checking works, but each organisation keeps its own record on its own rating scale. Detection tools rarely cover African languages: **67%** of studies use English data and **under 10%** target low-resource languages.

RESEARCH QUESTION

How can an integrated AI and blockchain framework enable tamper-resistant, transparent, multilingual monitoring and verification of election misinformation in South Africa?

Methodology

A four-stage pipeline

1. **Ingest:** claims enter an Apache Kafka stream that absorbs election-day surges.
2. **Triage:** AfroXLMR-large-76L (LoRA) flags likely misinformation with a confidence score.
3. **Verify:** every organisation except the submitter reviews; ⌊E/2⌋ + 1 matching votes finalise automatically, a split is DISPUTED.
4. **Record:** Hyperledger Fabric stores a SHA-256 hash and metadata; claim text stays off-chain in IPFS.

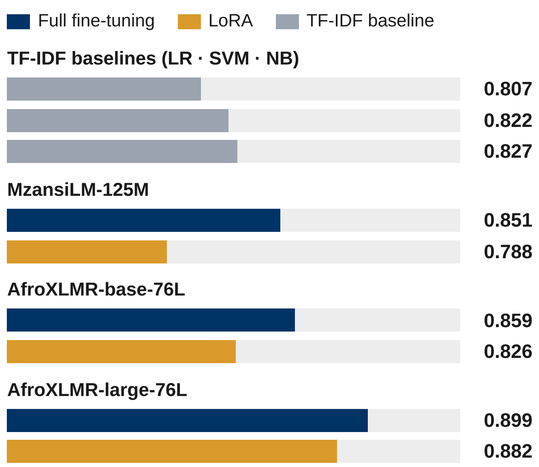
A synthetic multilingual corpus

No labelled election corpus existed in Sepedi or isiZulu. 5,000 scripted prompts asked Gemini 3.6 Flash for paired true and false statements, which were then machine-translated.

Language	Misinfo.	Accurate	Total
English	2,360	2,310	4,670
Sepedi	2,360	2,307	4,667
isiZulu	2,341	2,319	4,660
Total	**7,061**	**6,936**	**13,997**

Main results

Misinformation detection, test F1



Mean F1 (misinformation class) over three seeds on 1,393 held-out statements. Bars start at F1 = 0.70.

F1 by language, AfroXLMR-large-76L

Method	English	Sepedi	isiZulu
Full	0.926	0.885	0.889
LoRA	0.910	0.865	0.870

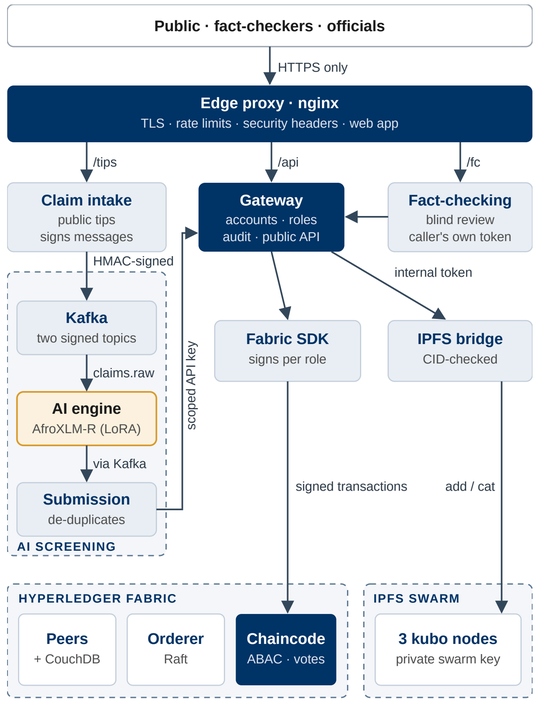
- Every AfroXLMR model beat the baselines; best single run reached **F1 0.906, AUC 0.963**.
- LoRA's gap to full fine-tuning shrinks with model size: **0.063 → 0.033 → 0.018**.
- Sepedi and isiZulu trail English by only about **0.04**.

WHY DEPLOY LORA?

Two of three full fine-tuning runs ended with validation loss 0.40–0.49 above training loss; every LoRA run stayed within 0.04, while training only 1.29% of parameters.

System architecture

A permissioned **Hyperledger Fabric** consortium with Raft ordering: no fees, deterministic finality, and roles enforced inside chaincode from X.509 certificate attributes.



One HTTPS edge is the only public entry point. The gateway signs each ledger transaction with the caller's role identity; claim text stays off-chain in a private IPFS swarm.

Official
submits, expires, admits members

Fact-checker
submits reports and fact-checks

Observer
read-only audit

POPIA by design

s 10 Only hash, CID, AI metadata and org ID on-chain.
s 13 Fixed schema; no free-form personal fields.
s 14 Claim text off-chain in IPFS, so deletable.
s 19 SHA-256, TLS channels and chaincode RBAC.

Key Highlights

13,997
statements in English, Sepedi and isiZulu

0.899
mean F1, AfroXLMR-large, full fine-tuning

0.882
mean F1 of the deployed LoRA model

750 / 750
feed workflows finalised and integrity-verified

~2.4 s
claim commitment at every size, 3 to 20 orgs

0 / 50
Caliper rounds with any failure, up to 2,000 tx

Conclusion

An Afrocentric encoder trained on translated synthetic data detects misinformation in Sepedi and isiZulu close to English. A Fabric consortium records joint verdicts tamper-evidently with no personal data on-chain; commitment stays stable while final verdicts slow as the required majority grows.

Limitations and next steps

Synthetic inputs, scripted reviews and one host. Next: real election posts, a multi-host deployment, a fact-checker web interface and per-user identities.

Acknowledgements

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Key references: Gondwe, Int. J. Press/Politics 2025 · Rananga et al., JISI 2024 · Alabi et al., COLING 2022 · Hu et al., ICLR 2022 · Androulaki et al., EuroSys 2018.

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